

Bitcoin Market Sentiment vs Trader Performance Analysis

Executive Summary

This project investigates the relationship between Bitcoin market sentiment and trader performance using historical trading data from Hyperliquid and the Bitcoin Fear & Greed Index. The objective is to identify how different market sentiment conditions influence profitability, trading behavior, and position sizing.

The analysis combines sentiment classifications (Extreme Fear, Fear, Neutral, Greed, and Extreme Greed) with trader performance metrics such as closed profit and loss (PnL) and position size. The findings reveal that market sentiment has a measurable impact on trading outcomes and can be used as an additional signal in trading decision-making.

Dataset Description

1. Bitcoin Fear & Greed Index

The sentiment dataset contains daily market sentiment classifications representing investor psychology and market conditions.

Key Fields:

- Date
- Classification (Extreme Fear, Fear, Neutral, Greed, Extreme Greed)

2. Hyperliquid Historical Trader Data

The trading dataset contains detailed records of executed trades.

Key Fields:

- Account
- Symbol
- Execution Price
- Side (Buy/Sell)
- Size
- Closed PnL
- Leverage
- Timestamp

The datasets were merged using the trading date to associate each trade with the corresponding market sentiment.

Methodology

1. Data Cleaning and Preprocessing
 2. Converted timestamps into a common date format.
 3. Removed missing and inconsistent values.
 4. Standardized sentiment labels.
 5. Data Integration
 6. Merged trader data with the Fear & Greed Index using trade dates.
 7. Exploratory Data Analysis
 8. Average profit by sentiment.
 9. Total profit by sentiment.
 10. Position size analysis.
 11. Feature importance evaluation.
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Results and Findings

1. Average Closed PnL by Sentiment

Extreme Greed produced the highest average profit per trade, followed by Fear and Greed conditions.

Key Observation:

- Extreme Greed $\approx$ Highest average profitability.
- Fear $\approx$ Second-highest average profitability.
- Neutral and Extreme Fear showed comparatively lower average returns.

Interpretation: Strong market momentum during Extreme Greed periods may create favorable trading opportunities, leading to higher profit per trade.

2. Total Profit by Sentiment

Fear generated the highest cumulative profit among all sentiment categories.

Key Observation:

- Fear $\approx$ Highest total profit.
- Extreme Greed $\approx$ Second highest total profit.

Interpretation: Although Extreme Greed generated larger profits per trade, Fear periods contributed more overall profit due to increased trading activity and larger position sizes.

3. Average Position Size by Sentiment

The largest average position sizes were observed during Fear conditions.

Key Observation:

- Fear $\approx$ Highest average position size.
- Extreme Greed $\approx$ Lowest average position size.

Interpretation: Traders appear to allocate more capital during Fear periods, possibly viewing market downturns as opportunities for larger positions.

4. Feature Importance Analysis

A feature importance model was used to evaluate factors influencing profitability.

Results:

- Position Size (USD): ~87%
- Sentiment: ~10%
- Trade Side (Buy/Sell): ~3%

Interpretation: Position size is the strongest determinant of profitability. Market sentiment contributes meaningfully but has a smaller influence compared to capital allocation decisions.

Business Insights

1. Market sentiment affects trader profitability and behavior.
 2. Extreme Greed conditions generate the highest average returns.
 3. Fear periods produce the highest cumulative profits.
 4. Position sizing is the most influential factor in determining trading outcomes.
 5. Sentiment indicators can be used as complementary signals in trading strategies.
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Recommendations

- Incorporate Fear & Greed Index signals into trading models.
 - Apply stricter risk controls during Extreme Greed conditions.
 - Focus on optimized position sizing to improve profitability.
 - Combine sentiment indicators with technical and quantitative signals for better decision-making.
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Conclusion

This analysis demonstrates a clear relationship between Bitcoin market sentiment and trader performance. While Extreme Greed conditions provide the highest average profit per trade, Fear periods contribute the largest overall profits. The findings also show that position size is the dominant driver of profitability, highlighting the importance of effective risk and capital management. Sentiment-based indicators can therefore serve as valuable supplementary tools for improving trading strategies and decision-making.